FEED-FORWARD NEURAL NETWORK

Layer L $(1 \leq L \leq L_0)$:

Input :

$$A^{(L-1)}$$

Weight :

$$W^{(L)} \ (m_L \times n_L)$$

Bias :

$$B^{(L)}$$

$$Z^{(L)} = W^{(L)} \cdot A^{(L-1)} + B^{(L)} \rightarrow Z_i^{(L)} = \sum_{j=1}^{n_L} W_{i,j}^{(L)} A_j^{(L-1)} + B_j^{(L)}$$

Output :

$$A^{(L)} = f_L(Z^{(L)}) \rightarrow A_i^{(L)} = f_L(Z_i^{(L)})$$

Loss :

Loss function :

$$C = \frac{1}{2} \sum_{i=1}^{n_{(L_0)}} \left(A_i^{(L_0)} - Y_i \right)^2$$

Gradient Layer L_0 :

We have the partial derivative of C respect to $W_{i,j}^{(L_0)}$:

$$\frac{\partial C}{\partial W_{i,j}^{(L_0)}} = \frac{\partial C}{\partial A_i^{(L_0)}} \cdot \frac{\partial A_i^{(L_0)}}{\partial Z_i^{(L_0)}} \cdot \frac{\partial Z_i^{(L_0)}}{\partial W_{i,j}^{(L_0)}}$$

Let

$$\delta_i^{(L_0)} = \frac{\partial C}{\partial A_i^{(L_0)}} \cdot \frac{\partial A_i^{(L_0)}}{\partial Z_i^{(L_0)}} = (A_i^{(L_0)} - Y_i) \cdot f'_{L_0} \left(Z_i^{(L_0)} \right)$$

And

$$\frac{\partial Z_i^{(L_0)}}{\partial W_{i,j}^{(L_0)}} = A_j^{(L_0-1)}$$

$$\Rightarrow \frac{\partial C}{\partial W_{i,j}^{(L_0)}} = \delta_i^{(L_0)} \cdot A_j^{(L_0-1)}$$

We have the partial derivative of C respect to $B_i^{(L_0)}$:

$$\frac{\partial C}{\partial B_i^{(L_0)}} = \frac{\partial C}{\partial A_i^{(L_0)}} \cdot \frac{\partial A_i^{(L_0)}}{\partial Z_i^{(L_0)}} \cdot \frac{\partial Z_i^{(L_0)}}{\partial B_i^{(L_0)}} = \delta_i^{(L_0)}$$

Gradient Layer $L_0 - 1$:

We have the partial derivative of C respect to $W_{i,j}^{(L_0-1)}$:

$$\frac{\partial C}{\partial W_{i,j}^{(L_0-1)}} = \sum_{k=1}^{m_{L_0}} \left(\frac{\partial C}{\partial A_k^{(L_0)}} \cdot \frac{\partial A_k^{(L_0)}}{\partial Z_k^{(L_0)}} \cdot \frac{\partial Z_k^{(L_0)}}{\partial A_i^{(L_0-1)}} \right) \cdot \left(\frac{\partial A_i^{(L_0-1)}}{\partial Z_i^{(L_0-1)}} \cdot \frac{\partial Z_i^{(L_0-1)}}{\partial W_{i,j}^{(L_0-1)}} \right)$$

As we defined above:

$$\delta_i^{(L_0)} = (A_i^{(L_0)} - Y_i) \cdot \frac{\partial A_i^{(L_0)}}{\partial Z_i^{(L_0)}} = \frac{\partial C}{\partial A_i^{(L_0)}} \cdot \frac{\partial A_i^{(L_0)}}{\partial Z_i^{(L_0)}}$$

and

$$W_{k,j}^{(L_0)} = \frac{\partial Z_i^{(L_0)}}{\partial A_i^{(L_0-1)}}$$

$$\Rightarrow \sum_{k=1}^{m_{L_0}} \left(\frac{\partial C}{\partial A_k^{(L_0)}} \cdot \frac{\partial A_k^{(L_0)}}{\partial Z_k^{(L_0)}} \cdot \frac{\partial Z_k^{(L_0)}}{\partial A_i^{(L_0-1)}} \right) = \sum_{k=1}^{m_{L_0}} \left(\delta_k^{(L_0)} \cdot W_{k,i}^{(L_0)} \right)$$

We have

$$f'_{L_0-1}\left(Z_i^{(L_0-1)}\right) = \frac{\partial A_i^{(L_0-1)}}{\partial Z_i^{(L_0-1)}}$$

and

$$A_j^{(L_0-2)} = \frac{\partial Z_i^{(L_0-1)}}{\partial W_{i,j}^{(L_0-1)}}$$

Let

$$\delta_i^{(L_0-1)} = \sum_{k=1}^{m_{L_0}} \left(\delta_k^{(L_0)} \cdot W_{k,i}^{(L_0)} \right) \cdot f'_{L_0-1}\left(Z_i^{(L_0-1)}\right)$$

Therefore :

$$\frac{\partial C}{\partial W_{i,j}^{(L_0-1)}} = \delta_i^{(L_0-1)} \cdot A_j^{(L_0-2)}$$

We have the partial derivative of C respect to $B_i^{(L_0-1)}$:

$$\frac{\partial C}{\partial B_i^{(L_0-1)}} = \sum_{k=1}^{m_{L_0}} \left(\frac{\partial C}{\partial A_k^{(L_0)}} \cdot \frac{\partial A_k^{(L_0)}}{\partial Z_k^{(L_0)}} \cdot \frac{\partial Z_k^{(L_0)}}{\partial A_i^{(L_0-1)}} \right) \cdot \left(\frac{\partial A_i^{(L_0-1)}}{\partial Z_i^{(L_0-1)}} \cdot \frac{\partial Z_i^{(L_0-1)}}{\partial B_i^{(L_0-1)}} \right) = \delta_i^{(L_0-1)}$$

Gradient of layer L $(1 \leq L \leq L_0)$:

Similarly to the above layer, we obtain :

$$\delta_i^{(L)} = \begin{cases} \left(A_i^{(L_0)} - Y_i\right) \cdot f'_{L_0}\left(Z_i^{(L_0)}\right), \ L = L_0 \\ \left(\sum_{k=1}^{m_{L+1}} \left(\delta_k^{(L+1)} \cdot W_{k,i}^{(L+1)}\right)\right) \cdot f'_L\left(Z_i^{(L)}\right), \ 1 \leq L < L_0 \end{cases}$$

and

$$\frac{\partial C}{\partial W_{i,j}^{(L)}} = \delta_i^{(L)} \cdot A_j^{(L-1)}$$

$$\frac{\partial C}{\partial B_i^{(L)}} = \delta_i^{(L)}$$

Matrix Form :

We denote some matrices as follows:
Activation matrix :

$$A^{(L)} = \begin{bmatrix} A_1^{(L)} \\ A_2^{(L)} \\ \cdots \\ A_{n_L}^{(L)} \end{bmatrix} \ (n_L \times 1)$$

Weight matrix :

$$W^{(L)} = \begin{bmatrix} W_{1,1}^{(L)} & W_{1,2}^{(L)} & \cdots & W_{1,n_L}^{(L)} \\ W_{2,1}^{(L)} & W_{2,2}^{(L)} & \cdots & W_{2,n_L}^{(L)} \\ \cdots & \cdots & \cdots & \cdots \\ W_{m_L,1}^{(L)} & W_{m_L,2}^{(L)} & \cdots & W_{m_L,n_L}^{(L)} \end{bmatrix} \ (m_L \times n_L)$$

Bias matrix :

$$B^{(L)} = \begin{bmatrix} B_1^{(L)} \\ B_2^{(L)} \\ \cdots \\ B_{m_L}^{(L)} \end{bmatrix} \ (m_L \times 1)$$

From these matrices, we derive :

$$Z^{(L)} = W^{(L)} \times A^{(L-1)} + B^{(L)} = \begin{bmatrix} Z_1^{(L)} \\ Z_2^{(L)} \\ \cdots \\ Z_{m_L}^{(L)} \end{bmatrix} \ (m_L \times 1)$$

We calculate $A^{(L)}$ from $Z^{(L)}$ as follows:

$$A^{(L)} = f_L\left(Z^{(L)}\right) = \begin{bmatrix} f_L\left(Z_1^{(L)}\right) \\ f_L\left(Z_2^{(L)}\right) \\ \cdots \\ f_L\left(Z_{m_L}^{(L)}\right) \end{bmatrix} \ (m_L \times 1)$$

Label matrix:

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \dots \\ Y_{m_{L_0}} \end{bmatrix} \; (m_{L_0} \times 1)$$

Let consider $\delta_i^{(L)}$:

- With $L = L_0$:

We have :

$$A^{(L_0)} - Y = \begin{bmatrix} A_1^{(L_0)} - Y_i \\ A_2^{(L_0)} - Y_i \\ \dots \\ A_{m_{L_0}}^{(L_0)} - Y_{m_{L_0}} \end{bmatrix} \; (m_{L_0} \times 1)$$

$$f'_L\left(Z^{(L_0)}\right) = \begin{bmatrix} f_L\left(Z_1^{(L_0)}\right) \\ f_L\left(Z_2^{(L_0)}\right) \\ \dots \\ f_L\left(Z_{m_{L_0}}^{(L_0)}\right) \end{bmatrix} \; (m_{L_0} \times 1)$$

From the formula of δ above, we obtain :

$$\delta^{(L_0)} = \left(A^{(L_0)} - Y\right) \odot f'_L\left(Z^{(L_0)}\right)$$

$$= \begin{bmatrix} \left(A_1^{(L_0)} - Y_1\right).f_L\left(Z_1^{(L_0)}\right) \\ \left(A_2^{(L_0)} - Y_2\right).f_L\left(Z_2^{(L_0)}\right) \\ \dots \\ \left(A_{m_{L_0}}^{(L_0)} - Y_i\right).f_L\left(Z_{m_{L_0}}^{(L_0)}\right) \end{bmatrix}$$

$$= \begin{bmatrix} \delta_1^{(L_0)} \\ \delta_2^{(L_0)} \\ \dots \\ \delta_{m_{L_0}}^{(L_0)} \end{bmatrix} \; (m_{L_0} \times 1)$$

- With $1 \leq L < L_0$:

We have :

$$\left(W^{(L+1)}\right)^T \times \delta^{(L+1)} = \begin{bmatrix} \sum_{k=1}^{m_{L+1}} \left(W_{k,1}^{(L+1)} \cdot \delta_k^{(L+1)}\right) \\ \sum_{k=1}^{m_{L+1}} \left(W_{k,2}^{(L+1)} \cdot \delta_k^{(L+1)}\right) \\ \dots \\ \sum_{k=1}^{m_{L+1}} \left(W_{k,n_{L+1}}^{(L+1)} \cdot \delta_k^{(L+1)}\right) \end{bmatrix} (n_{L+1} \times 1)$$

$$f'_L\left(Z^{(L)}\right) = \begin{bmatrix} f_L\left(Z_1^{(L)}\right) \\ f_L\left(Z_2^{(L)}\right) \\ \dots \\ f_L\left(Z_{m_L}^{(L)}\right) \end{bmatrix} \ (m_L \times 1)$$

Output of layer L ($A^{(L)}$) is the input of layer L + 1, thus $m_L = n_{L+1}$
Based on the formula of δ that we metioned above :

$$\left(\sum_{k=1}^{m_{L+1}} \left(\delta_k^{(L+1)} \cdot W_{k,i}^{(L+1)}\right)\right) \cdot f'_L\left(Z_i^{(L)}\right),\ 1 \leq L < L_0$$

We obtain δ matrix :

$$\delta^{(L)} = \left(\left(W^{(L+1)}\right)^T \times \delta^{(L+1)}\right) \odot f'_L\left(Z^{(L)}\right)$$

$$= \begin{bmatrix} \sum_{k=1}^{m_{L+1}} \left(W_{k,1}^{(L+1)} \cdot \delta_k^{(L+1)}\right) \cdot f_L\left(Z_1^{(L)}\right) \\ \sum_{k=1}^{m_{L+1}} \left(W_{k,2}^{(L+1)} \cdot \delta_k^{(L+1)}\right) \cdot f_L\left(Z_2^{(L)}\right) \\ \dots \\ \sum_{k=1}^{m_{L+1}} \left(W_{k,n_{L+1}}^{(L+1)} \cdot \delta_k^{(L+1)}\right) \cdot f_L\left(Z_{m_L}^{(L)}\right) \end{bmatrix} (m_L \times 1)$$

Conclusion :

Activation matrix :

$$A^{(L)} = \begin{bmatrix} A_1^{(L)} \\ A_2^{(L)} \\ \dots \\ A_{n_L}^{(L)} \end{bmatrix} \ (n_L \times 1)$$

Weight matrix :

$$W^{(L)} = \begin{bmatrix} W_{1,1}^{(L)} & W_{1,2}^{(L)} & \dots & W_{1,n_L}^{(L)} \\ W_{2,1}^{(L)} & W_{2,2}^{(L)} & \dots & W_{2,n_L}^{(L)} \\ \dots & \dots & \dots & \dots \\ W_{m_L,1}^{(L)} & W_{m_L,2}^{(L)} & \dots & W_{m_L,n_L}^{(L)} \end{bmatrix} \ (m_L \times n_L)$$

Bias matrix :

$$B^{(L)} = \begin{bmatrix} B_1^{(L)} \\ B_2^{(L)} \\ \dots \\ B_{m_L}^{(L)} \end{bmatrix} \ (m_L \times 1)$$

Z matrix :

$$Z^{(L)} = W^{(L)} \times A^{(L-1)} + B^{(L)}$$

Activation matrix from Z matrix :

$$A^{(L)} = f_L\left(Z^{(L)}\right) = \begin{bmatrix} f_L\left(Z_1^{(L)}\right) \\ f_L\left(Z_2^{(L)}\right) \\ \dots \\ f_L\left(Z_{m_L}^{(L)}\right) \end{bmatrix} \ (m_L \times 1)$$

Label matrix:

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \dots \\ Y_{m_{L_0}} \end{bmatrix} \ (m_{L_0} \times 1)$$

Delta matrix :

$$\delta^{(L)} = \begin{cases} \delta^{(L_0)} = \left(A^{(L_0)} - Y\right) \odot f'_{L_0}\left(Z^{(L_0)}\right), & L = L_0 \\ \delta^{(L)} = \left(\left(W^{(L+1)}\right)^T \times \delta^{(L+1)}\right) \odot f'_L\left(Z^{(L)}\right), & 1 \leq L < L_0 \end{cases}$$

Gradient for layer L:

$$\frac{\partial C}{\partial W^{(L)}} = \delta^{(L)} \times \left(A^{(L-1)}\right)^T$$

$$\frac{\partial C}{\partial B^{(L)}} = \delta^{(L)}$$

BATCH/MINI-BATCH TRAINING

Size definition:
- m_L : The number of neurons of layer L
- $n_L = m_{L-1}$: The size of input for layer L
- q : The number of sample for a training epoch

Input matrix:

$$A^{(L-1)} = \begin{bmatrix} a_{1,1}^{(L-1)} & a_{1,2}^{(L-1)} & \dots & a_{1,q}^{(L-1)} \\ a_{2,1}^{(L-1)} & a_{2,2}^{(L-1)} & \dots & a_{2,q}^{(L-1)} \\ \dots & \dots & \dots & \dots \\ a_{n_L,1}^{(L-1)} & a_{n_L,2}^{(L-1)} & \dots & a_{n_L,q}^{(L-1)} \end{bmatrix} \ (n_L \times q)$$

Weight matrix :

$$W^{(L)} = \begin{bmatrix} W_{1,1}^{(L)} & W_{1,2}^{(L)} & \dots & W_{1,n_L}^{(L)} \\ W_{2,1}^{(L)} & W_{2,2}^{(L)} & \dots & W_{2,n_L}^{(L)} \\ \dots & \dots & \dots & \dots \\ W_{m_L,1}^{(L)} & W_{m_L,2}^{(L)} & \dots & W_{m_L,n_L}^{(L)} \end{bmatrix} \ (m_L \times m_{L-1}) = (m_L \times n_L)$$

Bias matrix :

$$B^{(L)} = \begin{bmatrix} B_1^{(L)} \\ B_2^{(L)} \\ \dots \\ B_{m_L}^{(L)} \end{bmatrix} \ (m_L \times 1)$$

Z matrix :

$$Z^{(L)} = W^{(L)}A^{(L-1)} \ + B^{(L)} \quad (m_L \times q)$$

Activation matrix (Output):

$$A^{(L)} = f_L\left(Z^{(L)}\right) = \begin{bmatrix} a_{1,1}^{(L)} & a_{1,2}^{(L)} & \dots & a_{1,q}^{(L)} \\ a_{2,1}^{(L)} & a_{2,2}^{(L)} & \dots & a_{2,q}^{(L)} \\ \dots & \dots & \dots & \dots \\ a_{m_L,1}^{(L)} & a_{m_L,2}^{(L)} & \dots & a_{m_L,q}^{(L)} \end{bmatrix} (m_L \times q)$$

Label matrix:

$$Y = \begin{bmatrix} y_{1,1} & y_{1,2} & \dots & y_{1,q} \\ y_{2,1} & y_{2,2} & \dots & y_{2,q} \\ \dots & \dots & \dots & \dots \\ y_{m_{L_0},1} & y_{m_{L_0},2} & \dots & y_{m_{L_0},q} \end{bmatrix} (m_{L_0} \times q)$$

Delta matrix :

$$\delta^{(L)} = \begin{cases} \delta^{(L_0)} = \left(A^{(L_0)} - Y\right) \odot f'_{L_0}\left(Z^{(L_0)}\right), & L = L_0 \\ \delta^{(L)} = \left(\left(W^{(L+1)}\right)^T \times \delta^{(L+1)}\right) \odot f'_L\left(Z^{(L)}\right), & 1 \le L < L_0 \end{cases} \quad (m_L \times q)$$

If there is class weights:

$$\omega = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \vdots \\ \omega_{m_{L_0}} \end{bmatrix} (m_{L_0} \times 1)$$

Then delta matrix will be:

$$\delta^{(L)} = \begin{cases} \delta^{(L_0)} = \left(A^{(L_0)} - Y\right) \odot f'_{L_0}\left(Z^{(L_0)}\right) \odot \omega, & L = L_0 \\ \delta^{(L)} = \left(\left(W^{(L+1)}\right)^T \times \delta^{(L+1)}\right) \odot f'_L\left(Z^{(L)}\right), & 1 \le L < L_0 \end{cases} \quad (m_L \times q)$$

Gradient for layer L:

$$\frac{\partial C}{\partial W^{(L)}} = \frac{1}{q}\delta^{(L)} \times \left(A^{(L-1)}\right)^T$$

$$\frac{\partial C}{\partial B^{(L)}} = \frac{1}{q}\delta^{(L)} \times 1^{q\times 1}$$

CROSS ENTROPY LOSS

Layer L ($1 \leq L \leq L_0$):

Input :

$$A^{(L-1)} \quad (n_L \times 1)$$

Weight :

$$W^{(L)} \quad (m_L \times n_L)$$

Bias :

$$B^{(L)} \quad (m_L \times 1)$$

$$z_i^{(L)} = \sum_{j=1}^{n_L} w_{i,j}^{(L)} a_j^{(L-1)} + b_j^{(L)} \Rightarrow Z^{(L)} = W^{(L)} \cdot A^{(L-1)} + B^{(L)} \quad (m_L \times 1)$$

Class weight:

We choose the activation functions as follows :

$$f_L(z) = \begin{cases} ReLu(z), 1 \leq L \leq L_0 \\ \sigma(z), L = L_0 \end{cases}$$

+ ReLU function :

$$ReLU(z) = \begin{cases} z, z > 0 \\ 0, z \leq 0 \end{cases}$$

+ Softmax function :

$$\sigma(z_i) = \sigma_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Output :

$$a_i^{(L)} = f_L(z_i^{(L)}) \Rightarrow A^{(L)} = f_L(Z^{(L)}) \quad (m_L \times 1)$$

Cross Entroy Loss:

Class weight :

$$\Omega = [\omega_i] \quad (m_{L_0} \times 1)$$

Label :

$$Y \quad (m_{L_0} \times 1)$$

Loss function :

$$C = -\sum_{i=1}^{m_{L_0}} \omega_i y_i \ln\left(a_i^{(L_0)}\right)$$

Gradient Layer L_0 :

We have the partial derivative of C respect to $W_{i,j}^{(L_0)}$:

$$\frac{\partial C}{\partial w_{i,j}^{(L_0)}} = \sum_{k=1}^{m_{L_0}} \left(\frac{\partial C}{\partial a_k^{(L_0)}} \cdot \frac{\partial a_k^{(L_0)}}{\partial z_i^{(L_0)}} \right) \cdot \frac{\partial z_i^{(L_0)}}{\partial w_{i,j}^{(L_0)}}$$

$$\frac{\partial C}{\partial a_k^{(L_0)}} = \frac{-y_k \omega_k}{a_k^{(L_0)}}$$

$$\frac{\partial a_k^{(L_0)}}{\partial z_i^{(L_0)}} = \frac{\partial \sigma_k}{\partial z_i^{(L_0)}} = \begin{cases} a_i^{(L_0)}(1 - a_i^{(L_0)}), k = i \\ -a_i^{(L_0)} a_k^{(L_0)}, k \neq i \end{cases}$$

Let

$$\delta_i^{(L_0)} = \sum_{k=1}^{m_{L_0}} \left(\frac{\partial C}{\partial a_k^{(L_0)}} \cdot \frac{\partial a_k^{(L_0)}}{\partial z_i^{(L_0)}} \right) = \frac{-y_i \omega_i}{a_i^{(L_0)}} \cdot a_i^{(L_0)}(1 - a_i^{(L_0)}) + \sum_{k \neq i} \left(\frac{y_k \omega_k}{a_k^{(L_0)}} a_i^{(L_0)} a_k^{(L_0)} \right)$$

$$= -y_i \omega_i + \sum_{k=1}^{m_{L_0}} \left(\frac{y_k \omega_k}{a_k^{(L_0)}} a_i^{(L_0)} a_k^{(L_0)} \right) = -y_i \omega_i + \sum_{k=1}^{m_{L_0}} y_k \omega_k a_i^{(L_0)} = \left(a_i^{(L_0)} \sum_{k=1}^{m_{L_0}} y_k \omega_k \right) - y_i \omega_i$$

There is only one $y_k = 1$ in the label, thus:

$$\sum_{k=1}^{m_{L_0}} y_k = y_c = 1 \Rightarrow a_i^{(L_0)} \sum_{k=1}^{m_{L_0}} y_k \omega_k - y_i \omega_i = a_i^{(L_0)} \omega_c - y_i \omega_i$$

Therefore :

$$\delta_i^{(L_0)} = a_i^{(L_0)} \omega_c - y_i \omega_i$$

And

$$\frac{\partial z_i^{(L_0)}}{\partial w_{i,j}^{(L_0)}} = a_j^{(L_0 - 1)}$$

$$\Rightarrow \frac{\partial C}{\partial w_{i,j}^{(L_0)}} = \delta_i^{(L_0)} \cdot a_j^{(L_0 - 1)}$$

We have the partial derivative of C respect to $B_i^{(L_0)}$:

$$\frac{\partial C}{\partial B_i^{(L_0)}} = \frac{\partial C}{\partial A_i^{(L_0)}} \cdot \frac{\partial A_i^{(L_0)}}{\partial Z_i^{(L_0)}} \cdot \frac{\partial Z_i^{(L_0)}}{\partial B_i^{(L_0)}} = \delta_i^{(L_0)}$$

$$\Rightarrow \delta_i^{(L)} = \begin{cases} a_i^{(L_0)} \omega_c - y_i \omega_i, L = L_0 \\ \\ \left(\sum_{k=1}^{m_{L+1}} \left(\delta_k^{(L+1)} \cdot w_{k,i}^{(L+1)} \right) \right) \cdot f_L' \left(z_i^{(L)} \right), \ 1 \leq L < L_0 \end{cases}$$

Let:

$$\Omega = [\omega_{i,j}] \quad (m_L 0 \times q)$$
$$\Omega_c = [\omega_c] \quad (m_L 0 \times q)$$

Matrix Form :

$$\delta^{(L)} = \begin{cases} A^{(L_0)} \odot \Omega_c - Y \odot \Omega, & L = L_0 \\ \left(\left(W^{(L+1)}\right)^T \times \delta^{(L+1)}\right) \odot f'_L\left(Z^{(L)}\right), & 1 \le L < L_0 \end{cases} \qquad (m_L \times q)$$

$$\frac{\partial C}{\partial W^{(L)}} = \frac{1}{q}\delta^{(L)} \times \left(A^{(L-1)}\right)^T$$

$$\frac{\partial C}{\partial B^{(L)}} = \frac{1}{q}\delta^{(L)} \times 1^{q\times 1}$$

COVOLUTION NEURAL NETWORK

Filter Layer 1:

Input :

$$X^{(1)} = \begin{bmatrix} x^{(1)}_{1,1} & x^{(1)}_{1,2} & \dots & x^{(1)}_{1,9} \\ x^{(1)}_{2,1} & x^{(1)}_{2,2} & \dots & x^{(1)}_{2,9} \\ \dots & \dots & \dots & \dots \\ x^{(1)}_{9,1} & x^{(1)}_{9,2} & \dots & x^{(1)}_{9,9} \end{bmatrix} (9 \times 9)$$

Filter :

$$F = \begin{bmatrix} f^{(1)}_{1,1} & f^{(1)}_{1,2} & f^{(1)}_{1,3} \\ f^{(1)}_{2,1} & f^{(1)}_{2,2} & f^{(1)}_{2,3} \\ f^{(1)}_{3,1} & f^{(1)}_{3,2} & f^{(1)}_{3,3} \end{bmatrix} (3 \times 3)$$

Bias : b Output :

$$y_{1,1} = x_{1,1}f_{1,1}+x_{1,2}f_{1,2}+x_{1,3}f_{1,3}+x_{2,1}f_{2,1}+x_{2,2}f_{2,2}+x_{2,3}f_{2,3}+x_{3,1}f_{3,1}+x_{3,2}f_{3,2}+x_{3,3}f_{3,3}+b$$

$$= \sum_{p=1}^{3}\sum_{q=1}^{3} x_{1+p-1,1+q-1}f_{p,q}$$

$$\Rightarrow y_{i,j} = \sum_{p=1}^{3}\sum_{q=1}^{3} x_{i+m-1,j+q-1}f_{m,q} + b$$

$$Y^{(1)} = \begin{bmatrix} y_{1,1} & y_{1,2} & \dots & y_{1,7} \\ y_{2,1} & y_{2,2} & \dots & y_{2,7} \\ \dots & \dots & \dots & \dots \\ y_{7,1} & y_{7,2} & \dots & y_{7,7} \end{bmatrix}$$

$$\frac{\partial y_{1,1}}{\partial f_{1,1}} = x_{1,1}$$

$$\Rightarrow \begin{cases} \frac{\partial y_{i,j}}{\partial f_{p,q}} = x_{(i-1)+p,(j-1)+q} \\ \frac{\partial y_{i,j}}{\partial b} = 1 \end{cases}$$

Stride : s

$$y_{i,j} = \sum_{a=1}^{3} \sum_{b=1}^{3} x_{i \cdot s + a - 1, j \cdot s + b - 1} f_{a,b}$$

$$\Rightarrow \begin{cases} \frac{\partial y_{i,j}}{\partial f_{p,q}} = x_{(i-1) \cdot s + p, (j-1) \cdot s + q} \\ \frac{\partial y_{i,j}}{\partial b} = 1 \end{cases}$$